

"Please describe how you have prepared for your intended major, including your readiness to succeed in your upper-division courses once you enroll at the university."

wordcount : 340

While soldering a broken circuit board under a microscope, I noticed the chip at its center, a black square with copper paths running out of every side. What was in there? I scraped at it until a city of lines and layers showed through. I had been programming chips like it without once wondering what was inside. Everything that has prepared me for engineering at the intersection of hardware and software started there, taking me inward one layer at a time.

The first layer was code, and the Full Self-Driving club taught me something my courses couldn't: how software behaves in the physical world. The first time I tested our 1/10-scale kart, I was certain of my logic. I watched it drive full speed into a wall. After sifting through a nest of wires I hadn't dared to touch, I found the problem one layer down, in the kart's power distribution. I dug into its schematics until I could trace every step from code to signal to motion. I brought that same approach to our full-size autonomous kart, this time as the team's lead embedded systems engineer. Yet every chip on that kart, like the one under my microscope, raised a question the club couldn't answer: what does it take to build one?

To find out, I spent nine weeks last summer in ExSTENT, an NSF-funded program, doing cleanroom work most undergraduates don't touch until upper-division labs. When a layer we etched came out 200 nanometers too deep, my group assumed we'd slipped up. But every step matched the recipe, so I wasn't ready to blame us. I traced the problem through the etcher's diagnostics and into the chamber itself, where I found contamination. Upper-division labs will bring harder problems, and I can't wait to solve them.

By the end, I had seen how a chip is built, but not how one is designed: how someone decides where every line in that city goes. Upper-division courses are where I'll learn to draw those lines myself. Someday, I want to look at a chip under a microscope and find a city that's mine.